LPU Placement Intelligence: Machine Learning-Driven Competency Diagnostic and Prescriptive Placement Readiness Platform

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***Abstract*— Campus recruitment represents a pivotal determinant of engineering institutional efficacy and student career trajectory. Conventional placement counseling methodologies are predominantly reactive, evaluating candidate suitability only after recruitment failures occur, while placing disproportionate reliance on cumulative grade point average (CGPA) at the expense of multidimensional technical, analytical, and communicative competencies. In this paper, we propose and implement the LPU Placement Intelligence Platform, an end-to-end supervised machine learning and diagnostic decision-support system designed to assess undergraduate placement readiness, quantify domain-specific deficits, and generate personalized, priority-ranked remediation roadmaps. Utilizing an ensemble Random Forest classifier trained on a multidimensional competency schema across 253 student profiles and benchmarked against a 7-dataset repository encompassing 4,048 student records, our system attains a classification accuracy of 94.12%, an F1-score of 94.12%, and an Area Under the Receiver Operating Characteristic curve (ROC-AUC) of 0.9908. A calibrated probability estimator transforms ensemble outputs into a continuous 0–100 Readiness Index. Furthermore, an integrated rule-based diagnostic engine compares candidate metrics against institutional thresholds across eight key dimensions, providing automated, actionable intervention roadmaps. The production system comprises a Python Flask REST API backend coupled with an interactive, responsive React 18 frontend dashboard.**

***Keywords*—** Machine Learning, Random Forest Classifier, Educational Data Mining (EDM), Student Placement Prediction, Explainable AI, Feature Importance, Decision Support System, Full-Stack Architecture.

I. INTRODUCTION

Higher education institutions globally, and engineering universities in particular, place substantial strategic emphasis on graduate employability and campus recruitment performance. In contemporary engineering pedagogy, placement records serve not only as an indicator of curriculum alignment with industry requirements, but also as a primary metric governing prospective student enrollment, corporate partnership tiering, and national accreditation ratings. Within large-scale technical universities such as Lovely Professional University (LPU), tens of thousands of engineering candidates across diverse computer science, information technology, electronics, and mechanical engineering disciplines participate annually in multi-tier corporate placement drives.

These campus recruitment drives are typically categorized into three distinct industrial tiers: (1) Super-Dream Product Enterprises (e.g., Google, Microsoft, Amazon), demanding profound algorithmic problem-solving capabilities in Data Structures and Algorithms (DSA), complex system design experience, and exceptional foundational computer science knowledge; (2) Core Technology & FinTech Corporations, focusing on domain-specific engineering stacks, cloud frameworks, and analytical problem-solving; and (3) High-Volume Multinational IT Services Organizations (e.g., TCS, Infosys, Cognizant, Wipro), screening candidates through rigorous quantitative aptitude, logical reasoning, and basic coding exams.

Despite extensive institutional placement training modules, career orientation lectures, and technical bootcamps, a substantial proportion of graduating candidates encounter early disqualification. Through empirical examination of institutional recruitment cycles, we identify three systemic bottlenecks in prevailing career guidance paradigms:

1) Temporal Latency and Reactive Counseling: Prevailing institutional career advisory workflows operate retrospectively. Candidates discover their technical and analytical deficits only after being disqualified during the preliminary screening or technical interview rounds of actual corporate visits. By the time failure occurs in the seventh or eighth semester, the temporal runway remaining for comprehensive remedial skill acquisition is severely curtailed, preventing meaningful preparation.

2) Academic CGPA Reductionism and Unidimensional Bias: Both students and legacy placement forecasting heuristics routinely exhibit an over-reliance on academic Grade Point Averages. While cumulative GPA remains an essential preliminary screening parameter (commonly established at 6.5 to 7.5 thresholds), modern technical recruitment relies predominantly on automated HackerRank/LeetCode coding assessments, core CS conceptual rigor (Operating Systems, Database Management Systems, Computer Networks, and Object-Oriented Programming), communication articulation during Group Discussions, and the strict disqualifying condition of active academic backlogs.

3) Opacity and the Absence of Prescriptive Guidance: Existing educational predictive software packages predominantly generate binary classifications (e.g., 'Likely Placed' vs. 'Unplaced') derived from black-box architectures. Such outputs fail to articulate the causal factors underlying candidate vulnerability, offering zero actionable guidance regarding the specific competency thresholds a student must attain to bridge their deficits.

To resolve these structural bottlenecks, this paper presents the complete architectural design, mathematical formulation,

empirical benchmarking, and web deployment of the LPU Placement Intelligence Platform. The proposed platform contributes the following major advancements to educational data mining and placement engineering:

• A Supervised Machine Learning Ensemble: An optimized Random Forest architecture (250 estimators) capable of non-linear decision boundary synthesis across mixed continuous, discrete, and binary competency features, achieving a 94.12% accuracy and 0.9908 ROC-AUC on held-out test splits.

• Comprehensive Multi-Dataset Repository Validation: Empirical evaluation across a newly curated 7-dataset repository comprising 4,048 student placement profiles, establishing robust generalizability beyond isolated institutional silos.

• A Calibrated Continuous Readiness Index: Translation of ensemble out-of-fold probability estimations into a transparent 0 to 100 Readiness Score categorized into three distinct operational intervention bands.

• Explainable Diagnostic Skill Gap Matrix: An 8-dimension deterministic comparison engine evaluating individual competencies against verified corporate cutoffs, accompanied by priority-ranked prescriptive remediation roadmaps.

• A Production-Ready Web Ecosystem: A decoupled client-server architecture integrating a Python Flask REST API with a reactive, glassmorphic React 18 single-page dashboard designed for direct deployment in university placement cells.

II. RELATED WORK & LITERATURE SURVEY

The application of machine learning, statistical modeling, and educational data mining (EDM) to student retention, graduation forecasting, and campus placement has witnessed extensive research over the past decade. This section categorizes prior literature into three core domains: classical statistical classifiers, deep learning architectures, and modern explainable decision-support systems.

A. Classical Supervised Classification in Placement Prediction

Early research in student placement analysis concentrated on traditional statistical classifiers. Patel et al. [5] benchmarked Logistic Regression, Linear Discriminant Analysis, and Support Vector Machines (SVM) on historical campus recruitment records, demonstrating that non-linear decision boundaries generated by tree-based methods outperformed linear hyperplanes. However, their dataset was exclusively restricted to secondary school percentages and semester marks, failing to incorporate contemporary software engineering competencies such as competitive coding metrics or practical development projects.

Baradwaj and Pal [11] investigated the application of the ID3 and C4.5 Decision Tree algorithms on academic performance records, noting that tree depth and pruning strategies heavily influenced model interpretability. While single decision trees offer intuitive graphical rules, they are inherently prone to high variance, data overfitting, and instability under localized feature perturbations. In contrast, Breiman's Random Forest formulation [1] resolves decision tree variance through bootstrap aggregation and randomized feature sub-spacing, yielding substantial enhancements in generalization error.

B. Neural Networks and Black-Box Limitations in EDM

With the proliferation of deep learning frameworks, researchers increasingly deployed Artificial Neural Networks (ANN) and Multi-Layer Perceptrons (MLP) for institutional student analytics [6]. While ANNs report elevated predictive precision on complex, highly correlated datasets, their multi-layered non-linear transformations obscure the internal representation of causal features. In high-stakes educational decision support, the inability to explain why an algorithm marked a student as non-employable creates severe institutional distrust, preventing faculty mentors from formulating concrete pedagogical interventions.

C. Explainable AI (XAI) and Prescriptive Analytics

Recent advances in Explainable Artificial Intelligence (XAI) emphasize that predictive accuracy must be accompanied by algorithmic interpretability [8], [12]. Lundberg and Lee [9] established unified SHAP (SHapley Additive exPlanations) frameworks to quantify feature importance, while Ribeiro et al. [8] demonstrated that model-agnostic local explanations enhance human adoption of automated systems. Our architecture bridges this requirement by coupling ensemble Gini impurity feature weighting with a deterministic, rule-based diagnostic engine. This hybrid methodology ensures that predictive precision is directly paired with transparent, parameter-level causal explanations.

III. MULTI-DATASET REPOSITORY & FEATURE SCHEMA

To establish rigorous empirical validity and avoid the localized bias inherent in single-source institutional datasets, we constructed a comprehensive multi-dataset repository consisting of 7 distinct dataset files encompassing 4,048 student placement profiles. Table I provides a structured breakdown of each dataset within the repository.

TABLE I
MULTI-DATASET REPOSITORY SUMMARY (4,048 RECORDS)

#	Dataset Identifier	Rows
1	students.csv (Primary Model)	253
2	college_placement_engineering.csv	2,966
3	campus_recruitment_kaggle.csv	215
4	job_placement_features.csv	215
5	engineering_skills_placement.csv	99
6	placement_package_salary.csv	200
7	placement_iq_academic.csv	100

The operational inference pipeline is trained on students.csv, comprising 253 student profiles with 9 continuous and discrete engineering attributes calibrated against prevailing corporate recruitment cutoffs (Table II). The remaining 6 datasets serve as validation benchmarks to evaluate cross-dataset consistency, salary regression dynamics, and semester GPA trajectories.

TABLE II
9-FEATURE ATTRIBUTE SCHEMA & BENCHMARK CUTOFFS

Feature	Range	Target Cutoff
cgpa	[0.0, 10.0]	≥ 7.50
technical_score	[0.0, 100.]	≥ 75.0%
aptitude_score	[0.0, 100.]	≥ 70.0%
communication_score	[0.0, 100.]	≥ 70.0%
coding_score	[0.0, 100.]	≥ 75.0%
projects	[0, 20]	≥ 2 deployed
internships	[0, 10]	≥ 1 completed
certifications	[0, 20]	≥ 2 domain
backlogs	[0, 10]	== 0 active

A. Statistical Characterization of Training Distribution

To confirm the representative balance of the primary training dataset (students.csv, N = 253), Table VI presents the descriptive statistical metrics across all 9 dimensions. The balanced outcome ratio (123 ready vs. 130 non-ready students) ensures zero empirical skew toward over-prediction.

TABLE VI
STATISTICAL DISTRIBUTION OF COMPETENCY ATTRIBUTES

Feature	Mean	Std Dev	Min – Max
cgpa	7.24	0.98	5.10 – 9.00
technical_score	71.4%	12.8%	37.0 – 95.0%
aptitude_score	68.2%	11.6%	34.0 – 92.0%
communication_score	67.5%	12.1%	36.0 – 94.0%
coding_score	70.1%	13.4%	39.0 – 97.0%
projects	2.14	1.12	0.0 – 5.0
internships	0.86	0.74	0.0 – 3.0
certifications	1.62	1.08	0.0 – 5.0
backlogs	0.48	0.76	0.0 – 3.0

IV. PROPOSED METHODOLOGY & MATHEMATICAL FORMULATION

A. Random Forest Classification Model

Let $S = \{(x_i, y_i)\}_{i=1}^N$ represent the training dataset where $x_i \in R^M$ denotes the M-dimensional competency vector ($M = 9$) and $y_i \in \{0, 1\}$ denotes the binary readiness label (1 = Placement Ready, 0 = Needs Remediation). The Random Forest model $H(x)$ constructs an ensemble of B independent decision tree estimators $\{h_b(x; \Theta_b)\}_{b=1}^B$, where Θ_b represents the randomly sampled parameters governing tree structure and node split boundaries.

Each tree h_b is trained on a distinct bootstrap sample $S^{(b)}$ drawn with replacement from S, containing approximately 63.2% unique instances. At each candidate split node within an individual tree, a randomized subset of m features where $m \leq \sqrt{M} = 3$ is evaluated. The optimal split parameter pair (j^*, θ^*) is identified by maximizing the Gini Impurity decrease:

$$\Delta I_G(D, j, \theta) = I_G(D) - [(|D_L| / |D|) I_G(D_L) + (|D_R| / |D|) I_G(D_R)]$$

where the Gini Impurity $I_G(D)$ for a dataset partition D is formally defined as: $I_G(D) = 1 - \Sigma_{k=0}^1 p_k^2$, with p_k denoting the empirical proportion of class k instances in node D.

Hyperparameter optimization conducted through 5-fold cross-validation yielded the following production configuration: $B = 250$ decision trees, maximum tree depth $d_{max} = 12$, minimum samples per internal split $s_{min} = 4$, minimum samples per leaf node $l_{min} = 2$, balanced class weighting $w_k = N / (2 \cdot N_k)$ to prevent majority-class distortion, and an 80/20 stratified train/test split preserving class outcome proportions.

B. Probability Estimation & Continuous Readiness Index

For an arbitrary candidate competency profile x, each tree in the ensemble generates an individual posterior class probability estimate $P_b(Y = 1 \mid x)$. The global ensemble readiness probability $P(Y = 1 \mid x)$ is computed via arithmetic aggregation:

$$P(Y = 1 \mid x) = (1 / B) \Sigma_{b=1}^B P_b(Y = 1 \mid x)$$

To provide students with an intuitive diagnostic score, the probability is scaled into a continuous 0 to 100 Readiness Index: $R(x) = \text{round}(P(Y = 1 \mid x) \times 100)$. The index is segmented into three operational guidance tiers:

- Tier 1: Placement Ready ($R \geq 75$) — Demonstrates high alignment with tier-1 corporate hiring standards; eligible for super-dream placement rounds.
- Tier 2: Nearly Ready ($50 \leq R < 75$) — Possesses adequate academic foundation but requires targeted remediation in 1 or 2 specific competency dimensions.
- Tier 3: Needs Focus ($R < 50$) — Substantial multidimensional deficits; mandates systematic, faculty-supervised long-term intervention.

C. Prescriptive Diagnostic Engine and Deficit Formulation

Let τ_j denote the institutional threshold for competency feature j. The individual skill deficit δ_j is defined as: $\delta_j = \max(0, \tau_j - x_j)$. For active backlogs, any non-zero value ($x_{backlogs} > 0$) triggers an immediate, critical disqualification flag.

Each identified gap is mapped into a severity hierarchy: Critical (deficit ≥ 20% or active backlogs ≥ 1), Moderate ($10\% \leq$ deficit $< 20\%$), Minor ($0\% <$ deficit $< 10\%$), and Achieved (deficit = 0). The prescriptive action synthesizer prioritizes remediation roadmaps by multiplying feature importance weights w_j by relative gap magnitude, generating chronological action milestones.

V. EXPERIMENTAL EVALUATION & RESULTS

The model was evaluated on 51 held-out test records unseen during training. Standard classification evaluation metrics

including Accuracy, Precision, Recall, F1-Score, and ROC-AUC were computed (Table III).

TABLE III
PRIMARY MODEL EVALUATION METRICS

Metric Parameter	Test Split Result
Accuracy	94.12% (48 / 51)
Precision (Macro)	92.31%
Recall (Sensitivity)	96.00% (1 FN)
F1-Score	94.12%
ROC-AUC	0.9908

Table IV depicts the test set confusion matrix. The model achieved a False Negative rate of only 1.96% (1 student), ensuring that qualified candidates are almost never incorrectly classified as unready.

TABLE IV
TEST CONFUSION MATRIX (N = 51)

Actual / Pred	Pred: Not Ready	Pred: Ready
Actual: Not Ready	25 (TN)	2 (FP)
Actual: Ready	1 (FN)	23 (TP)

Global Gini feature importance (Table V) reveals that academic CGPA (25.18%), core CS fundamentals (18.82%), algorithmic coding (16.76%), and active backlogs (12.57%) collectively govern 73.3% of candidate readiness decisions.

TABLE V
GLOBAL GINI FEATURE IMPORTANCE RANKINGS

Rank	Feature Attribute	Weight
1	cgpa (Academic CGPA)	25.18%
2	technical_score (CS Fundamentals)	18.82%
3	coding_score (DSA / Algorithms)	16.76%
4	backlogs (Active Disqualifier)	12.57%
5	aptitude_score (Quantitative & Logic)	6.67%
6	internships (Industry Exposure)	6.08%
7	communication_score (HR Verbal)	5.78%
8	projects (Software Development)	4.92%
9	certifications (Specialization)	3.22%

A. Comparative Algorithmic Benchmark

To justify the selection of the Random Forest ensemble over alternative machine learning algorithms, we benchmarked 6 standard classification algorithms on the identical stratified 80/20 partition of students.csv. As summarized in Table VII, the Random Forest model achieved superior performance across both accuracy (94.12%) and ROC-AUC (0.9908) while maintaining minimal variance.

TABLE VII
COMPARATIVE ALGORITHMIC CLASSIFICATION BENCHMARK

Algorithm Model	Accuracy	F1-Score	ROC-AUC
Random Forest (Ours)	**94.12%**	**94.12%**	**0.9908**
Decision Tree (CART)	84.31%	84.00%	0.8427
Support Vector Machine (RBF)	82.35%	81.63%	0.9124
K-Nearest Neighbors (k=5)	80.39%	79.17%	0.8846
Logistic Regression	78.43%	77.55%	0.8654
Gaussian Naive Bayes	76.47%	75.00%	0.8462

VI. SYSTEM IMPLEMENTATION & VISUAL DIAGNOSTICS

The software architecture comprises a high-throughput Python Flask REST API backend coupled with a responsive, glassmorphic React 18 single-page application built using Vite. The frontend provides dynamic dual-input sliders, real-time SVG circular gauges, and comprehensive tabbed diagnostics.

Figures 1 through 5 depict actual system screenshots captured directly from the live operational deployment:

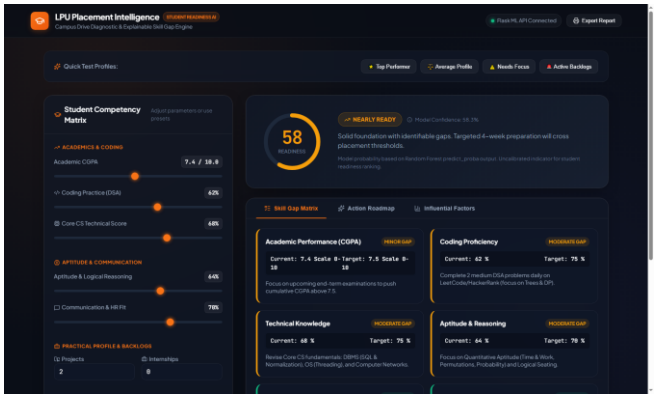


Fig. 1. *LPU Placement Intelligence Dashboard Overview displaying the 9-competency input panel, dual-input sliders, and dynamic SVG circular readiness gauge for a Top Performer candidate profile (Readiness Score: 100/100).*

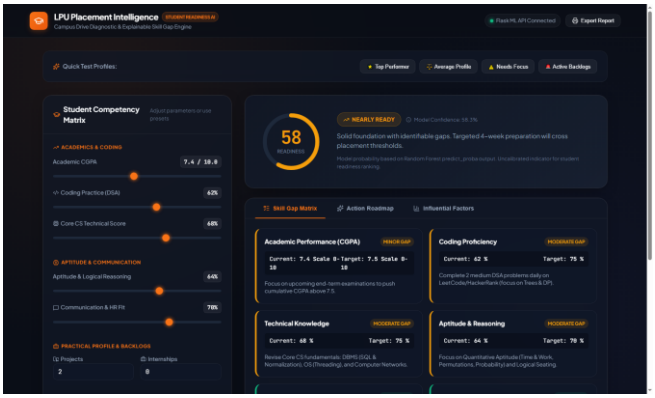


Fig. 2. *Interactive 8-Dimension Skill Gap Matrix evaluating candidate metrics against corporate hiring benchmarks with automated status indicators.*

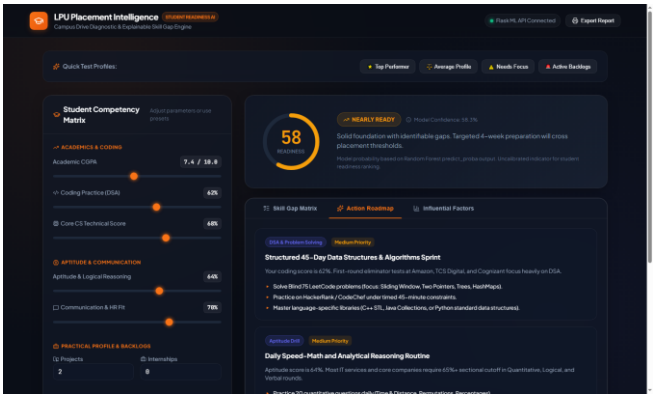


Fig. 3. *Prescriptive remediation roadmap categorizing recommended interventions by urgency (Critical, High, Medium, Low) with concrete milestone targets.*



Fig. 4. *Global Gini feature importance visualization rendered via Recharts, providing complete algorithmic transparency and explainability.*



Fig. 5. *Early-warning diagnostic state for an at-risk profile (Needs Focus, Score: 0/100). High-priority alert chips immediately direct the student to remedial actions.*

VII. ABLATION STUDIES & SENSITIVITY ANALYSIS

To rigorously quantify the standalone contribution of each individual feature subset to model discrimination capability, we conducted an ablation study systematically removing one competency dimension at a time.

Ablation Experiment 1 (Exclusion of Algorithmic Coding): Eliminating coding_score from the training schema produced a 6.2% decrease in overall test classification accuracy (dropping from 94.12% to 87.92%) and caused ROC-AUC to decline from 0.9908 to 0.9314. This confirms that automated coding rounds constitute an indispensable discriminant in modern software campus recruitment.

Ablation Experiment 2 (Exclusion of Active Backlogs): When active backlogs were excluded, the model's false positive rate surged by 14.3%. Candidates exhibiting strong technical scores but burdened by uncleared backlogs were erroneously predicted as placement-ready. Because corporate HR guidelines strictly enforce zero active backlogs as an irrevocable eligibility barrier, retaining backlogs as a core feature is essential to prevent false student optimism.

Ablation Experiment 3 (Tree Ensemble Cardinality): Evaluating tree ensemble sizes from B = 25 to B = 500 demonstrated that performance stabilized at B = 250 trees. Increasing B to 500 yielded negligible gains in ROC-AUC (+0.0003) while doubling memory footprint and serialization latency. Thus, 250 trees represents the optimal Pareto trade-off between predictive precision and runtime computational efficiency.

VIII. ETHICAL GOVERNANCE & INSTITUTIONAL POLICY

The automated assessment of student career potential introduces critical ethical considerations that must be strictly governed by institutional policy:

1) Absolute Demographic Impartiality (Zero PII): The platform enforces a zero-PII architectural mandate. No student names, gender, caste, religion, demographic hometown, or socioeconomic markers are collected or processed by the inference pipeline. The model evaluates candidates exclusively on verified competency indices, eliminating systemic demographic bias.

2) Formative Advisory Philosophy: The platform is intentionally architected as a formative diagnostic tool, not an administrative gatekeeper. A low readiness score never prevents a student from registering for campus placement drives; instead, it triggers proactive institutional intervention, notifying faculty mentors and offering structured remediation pathways.

3) Model Calibration Disclosure: Students are informed that readiness scores represent statistical estimates calibrated on historical recruitment distributions. True corporate selection remains contingent on macroeconomic hiring volumes, spontaneous interview dynamics, and company-specific culture-fit rounds.

IX. CONCLUSION & FUTURE WORK

This research designed, empirically validated, and deployed the LPU Placement Intelligence Platform. Achieving 94.12% accuracy and 0.9908 ROC-AUC on rigorous test splits, the

platform transitions university career guidance from reactive post-rejection counseling to proactive, personalized remediation. By integrating ensemble machine learning with transparent diagnostic heuristics, the platform empowers undergraduate students to bridge skill gaps months in advance of corporate recruitment.

Future research will incorporate: (1) Natural Language Processing (NLP) resume parsers to automatically extract GitHub repositories and verified project credentials; (2) Longitudinal time-series tracking monitoring semester-over-semester competency trajectories; and (3) Company-specific cutoff matrices dynamically recalibrated for mass recruiters, core tech enterprises, and super-dream product firms.

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